

MIT 18.02SC Multivariable Calculus Cheat sheet 1

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1 Functions, Traces, and Visualization

A function $z = f(x, y)$ involves two **independent variables** (x, y) and one **dependent variable** (z) , requiring **three dimensions** for graphing.

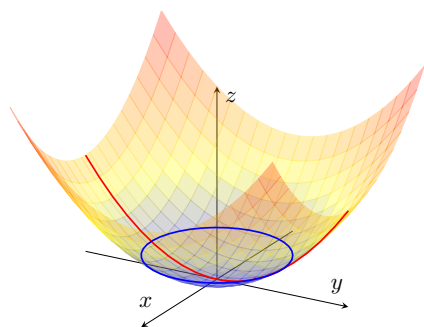
Geometric Definitions

- **Traces:** Curves found by setting one variable to a **constant**. For $z = x^2 + y^2$, the yz -trace (where $x = 0$) is an **upward-pointing parabola**.
- **Level Curves:** Curves in the xy -plane where $f(x, y) = c$. These are "pulled out" to their correct height to form the 3D graph.
- **Contour Plot:** A plot of level curves labeled with their z -values. A **mountain pass** is represented by a level curve that crosses itself.

Gallery of Graphs

- **Ellipsoid:** $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.
- **Elliptic Paraboloid:** $z = ax^2 + by^2$.
- **Hyperbolic Paraboloid:** $z = by^2 - ax^2$ (Saddle shape).
- **Elliptic Cone:** $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$.

Trace and Level Curves of $z = x^2 + y^2$



2 Partial Derivatives and Smoothness

The **partial derivative** is the ordinary derivative of a **partial function** (e.g., $w = f(x, y_0)$ where y is fixed).

- **Interpretation:** $\frac{\partial w}{\partial x}$ gives the **rate of change** of w with respect to x alone. Geometrically, it is the **slope** of the x -trace curve at point P .
- **Smoothness Hypothesis:** $f(x, y)$ is **smooth** at (x_0, y_0) if f_x and f_y are **continuous** in a rectangle centered there.
- **The Pathological Cone:** $w = \sqrt{x^2 + y^2}$ is **not smooth** at the origin because the graph has a **sharp vertex**; the partials are **undefined** there, and **no tangent plane** exists.

3 The Tangent Approximation and Sensitivity

The Tangent Plane

The **tangent plane** at (x_0, y_0, w_0) contains the **tangent lines** to the graphs of the two partial functions. **Equation:**

$$w - w_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Approximation formula (Δ notation)

For small changes $\Delta x, \Delta y$:

$$\Delta w \approx f_x \Delta x + f_y \Delta y$$

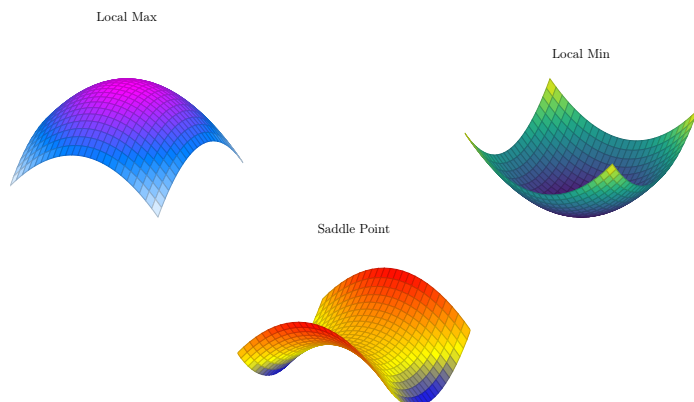
The right side is the **linearization** of $f(x, y)$ —the **best linear approximation** near (x_0, y_0) .

Example: Sensitivity Principle The numerical value of a function is most sensitive to the variable whose partial derivative has the **largest absolute value** at that point.

- For box volume $V = abc$ at $(1, 2, 3)$: $\Delta V \approx 6\Delta a + 3\Delta b + 2\Delta c$.
- Result: Volume is **most sensitive to side a** (the shortest side).

4 Optimization and Critical Points

Extrema (maxima/minima) occur at **critical points** where $f_x = 0$ and $f_y = 0$. At these points, the **tangent plane is horizontal**.



Example: Minimum Material Box Find dimensions for $V = xyz = 3$ with **double thick sides** ($4yz$), **triple thick bottom** ($3xy$), and **single thick front/back** ($2xz$).

1. $w = 4yz + 2xz + 3xy$. Substitute $z = 3/xy$:
 $w = \frac{12}{x} + \frac{6}{y} + 3xy$.
2. Set **partials to zero**: $w_x = -12/x^2 + 3y = 0$
and $w_y = -6/y^2 + 3x = 0$.
3. **Optimal Solution**: $x = 2, y = 1, z = 1.5$.

5 Method of Least Squares

To find the **least-squares line** $y = ax + b$ that "best" passes through data points, minimize the **sum of squared deviations**: $D = \sum [y_i - (ax_i + b)]^2$.

Normal Equations: Setting $\frac{\partial D}{\partial a} = 0$ and $\frac{\partial D}{\partial b} = 0$ leads to:

1. $(\sum x_i^2)a + (\sum x_i)b = \sum x_i y_i$
2. $(\sum x_i)a + nb = \sum y_i$

This method generalizes to **higher-dimensional data**, such as fitting $z = a_1 + a_2x + a_3y$ to points (x_i, y_i, z_i) .